

Introduction

- Dairy farm-level energy optimisation remains largely unexplored.
- Previous dairy energy studies relied on retrospective data analysis, lacking active energy management.
- Digital Twins (DTs) provide dynamic, virtual representations of physical systems that empowers data-driven decision-making.
- The use of DTs has not been well established within agriculture yet [1].
- To **reduce carbon emissions and operational costs**, this project establishes a framework for an AI-powered Digital Twin to **benchmark operational efficiency** using the watt-hours per litre of milk (WhLm⁻¹) metric; and **evaluate renewable energy (RE) viability**.

Study Design & Site Selection

- In collaboration with AgriSearch, a systematic methodology selected six representative sites from 73 surveyed farms.
- Single and three-phase farms selected for robust, real-world framework validation.

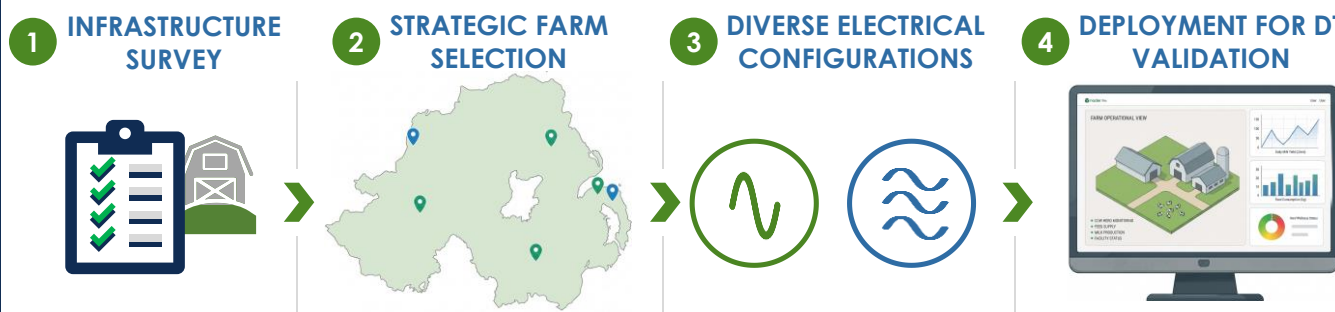


Fig 1. Workflow for Site Selection and DT Deployment

Framework Architecture

- Framework operates on an IoT-driven architecture, and automated scheduled tasks for continuous data ingestion.
- DT system is powered by a software application and a database, enabling data synchronisation and active performance monitoring across the farm network.

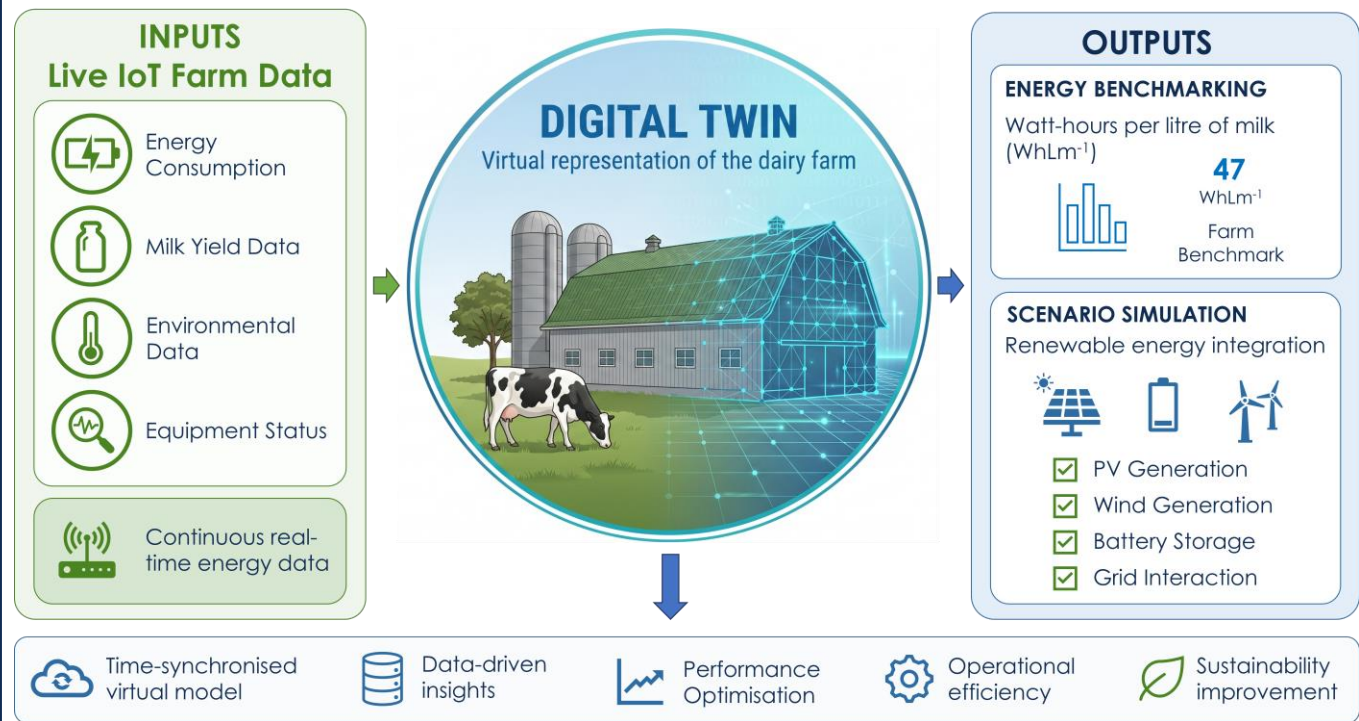


Fig 2. Proposed Architectural Structure for Energy Digital Twin

Methodology: Hardware Deployment

- To maximise aggregate load visibility, a combination of data logging equipment was deployed across both single and three-phase farm electrical configurations.

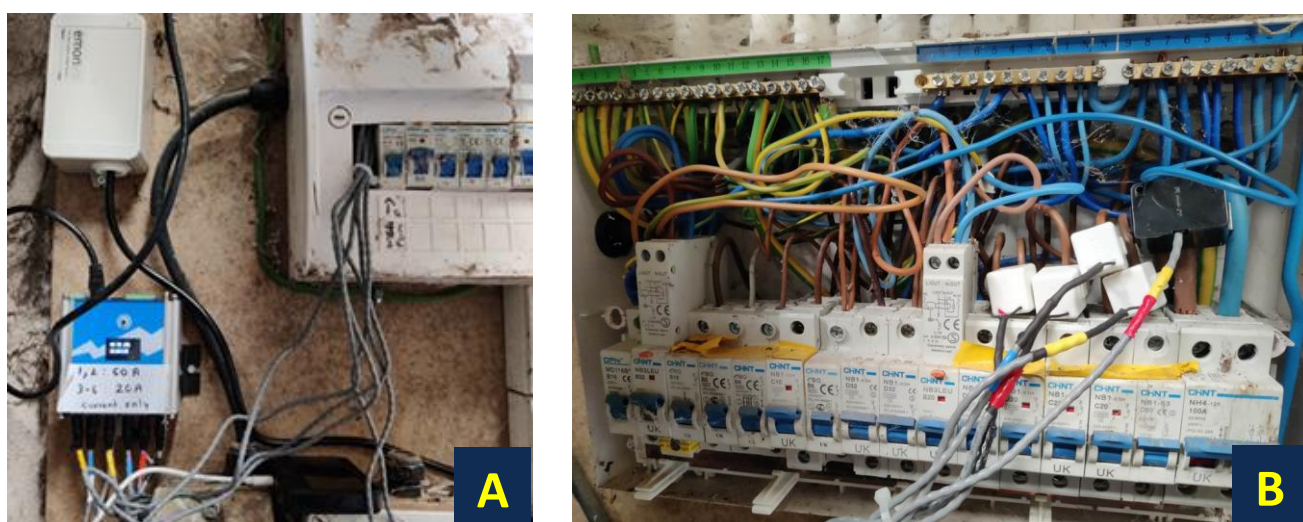


Fig 3. Energy monitoring meter (A) and CT clamp installation (B)

Methodology: DT Implementation

Real-Time Telemetry & Reporting:

- Integrating DT web application with live telemetry ensures continuous farm-level energy visibility.
- Disaggregating mains consumption enables granular load profiling and component-level benchmarking.

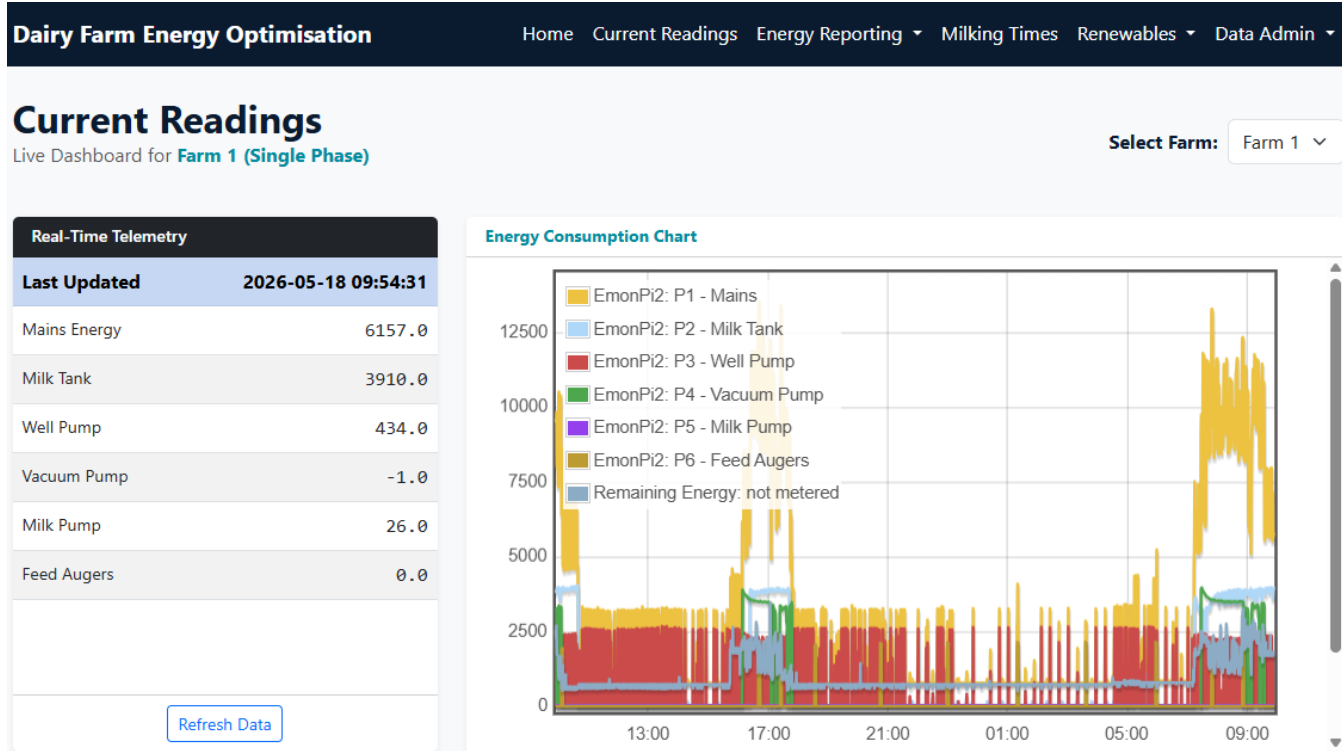


Fig 4. Real-time energy readings and 24-hour energy chart

- Energy reporting functionality facilitates daily, weekly, and monthly profiling of significant energy consumers.

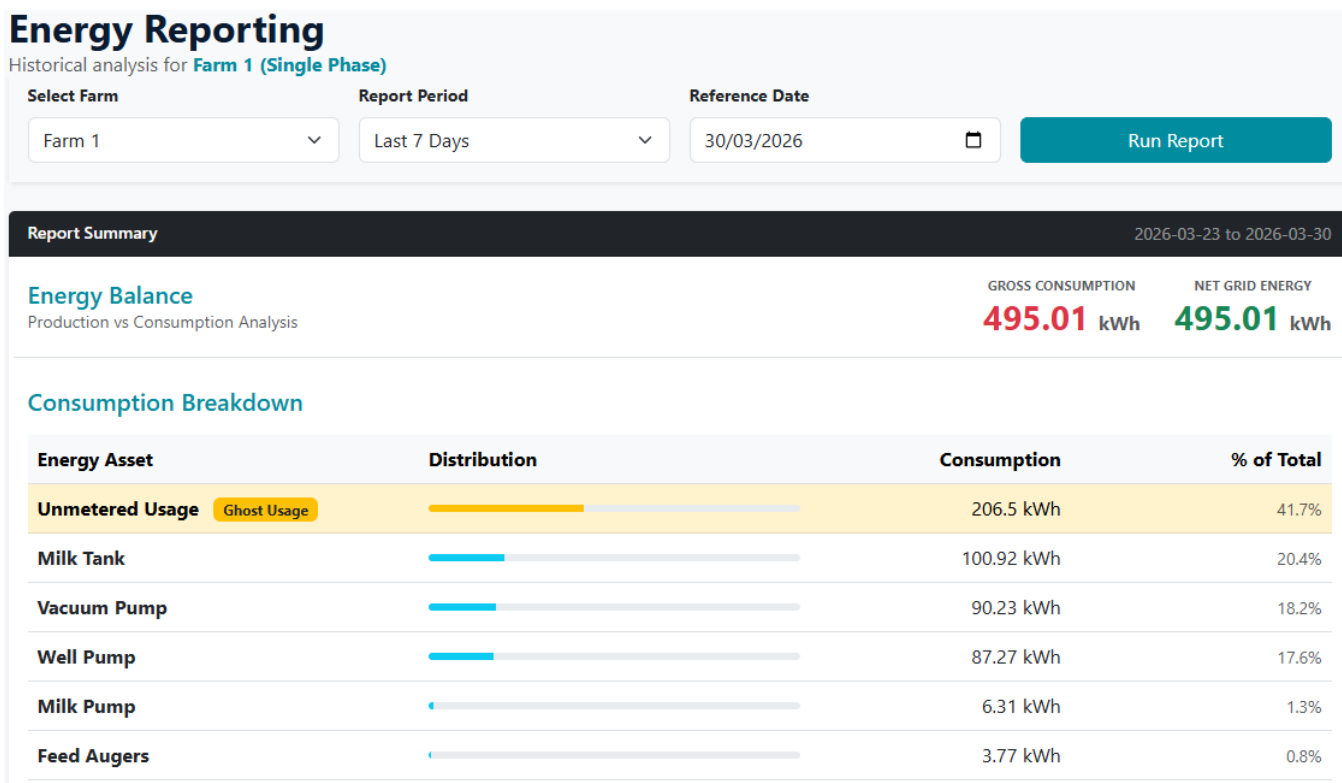


Fig 5. Farm level energy consumption reporting functionality

- Cross-farm reporting enables comparative benchmarking of both mains and individual asset consumption.

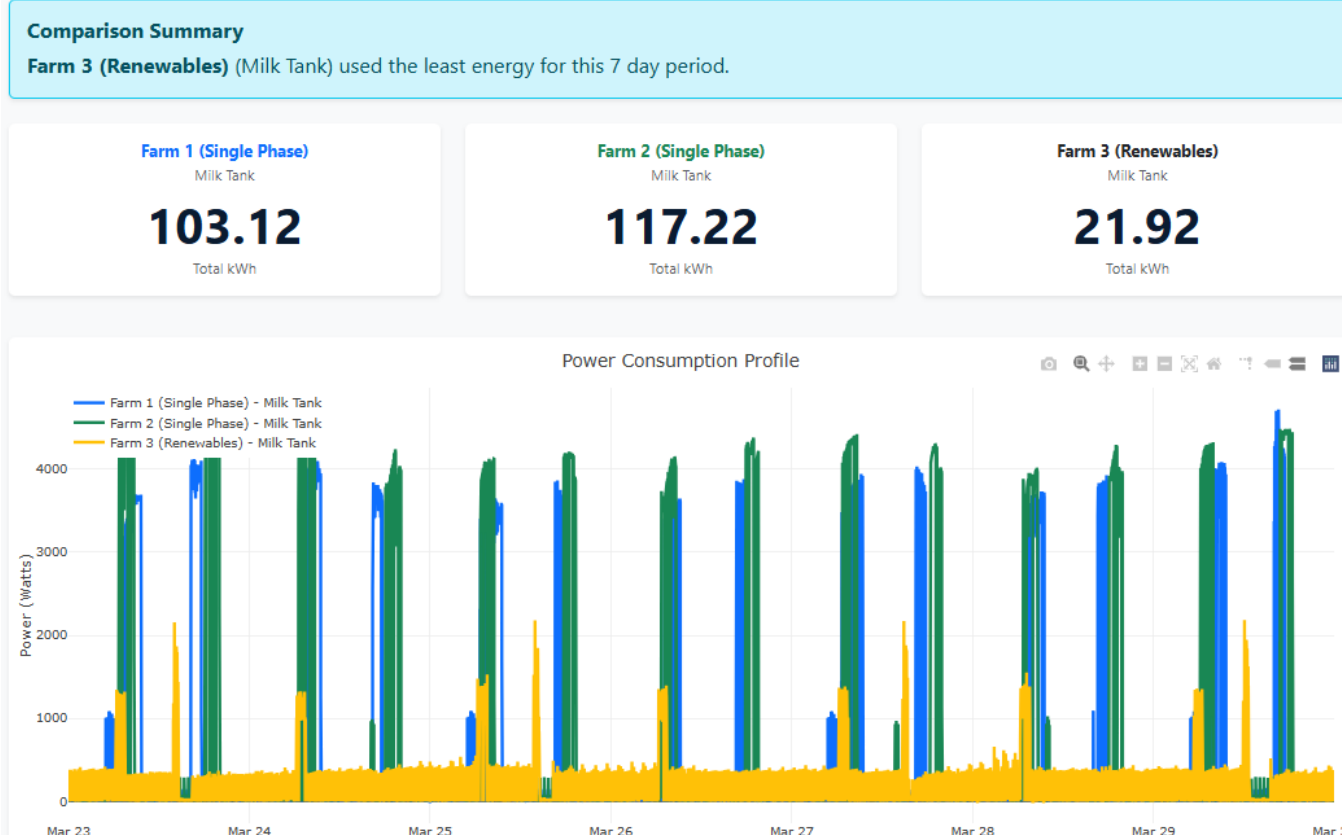


Fig 6. Weekly Milk Tank energy consumption compared on 3 farms

- Milking duration reporting facilitates comparative analysis of labour costs.
- Identifying anomalous pump usage during wash cycles validates the framework's detection and alerting capabilities.

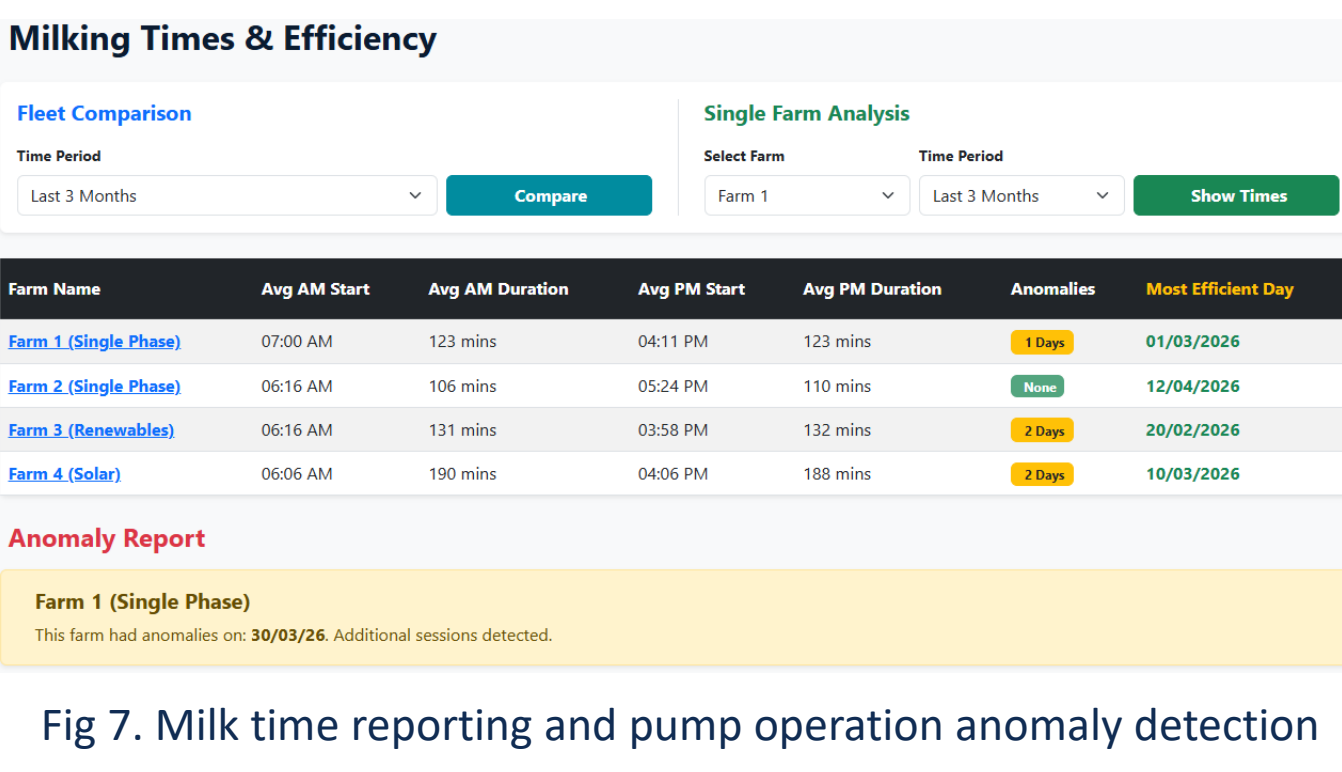


Fig 7. Milk time reporting and pump operation anomaly detection

References:
[1] C. Pylaniadis, S. Osinga, and I. N. Athanasiadis, 'Introducing digital twins to agriculture', Comput. Electron. Agric., vol. 184, p. 105942, May 2021, doi: 10.1016/j.compag.2020.105942.
[2] 'DSSE: Decision Support System for Energy use in Dairy Production', Teagasc, SEAI Project RDD/00117-Project Report, Nov. 2021. Accessed: Dec. 17, 2024. [Online]. Available: https://t-stor.teagasc.ie/handle/11019/2636
[3] P. Shine, J. Upton, P. Sefeedpari, and M. D. Murphy, 'Energy Consumption on Dairy Farms: A Review of Monitoring, Prediction Modelling, and Analyses', Energies, vol. 13, no. 5, p. 1288, Mar. 2020, doi: 10.3390/en13051288.

Results: Simulation & Benchmarking

- RE Integration:** Machine Learning scenario modelling simulates solar and wind integration to quantify potential energy, financial, and environmental benefits.

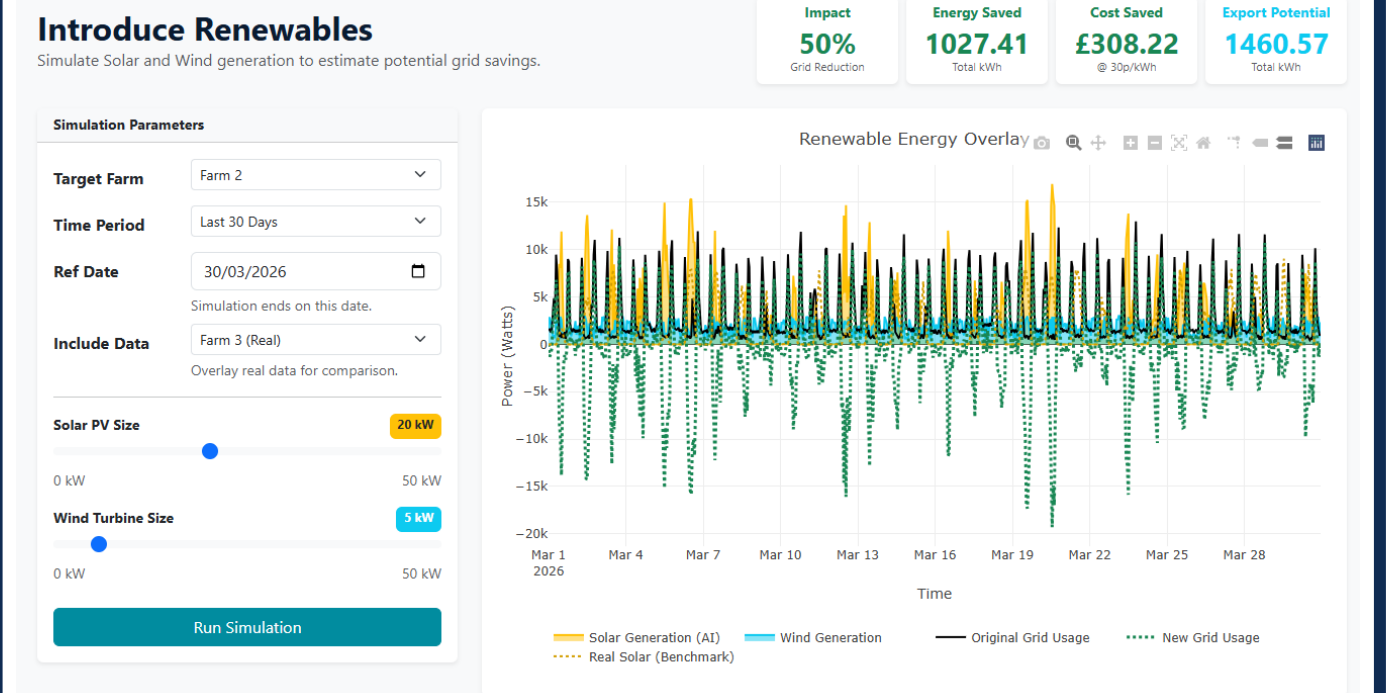


Fig 8. Simulated environmental and cost impact of RE integration

- Retrospective validation of the solar forecasting model against empirical data demonstrated 75% prediction accuracy at a selected farm.

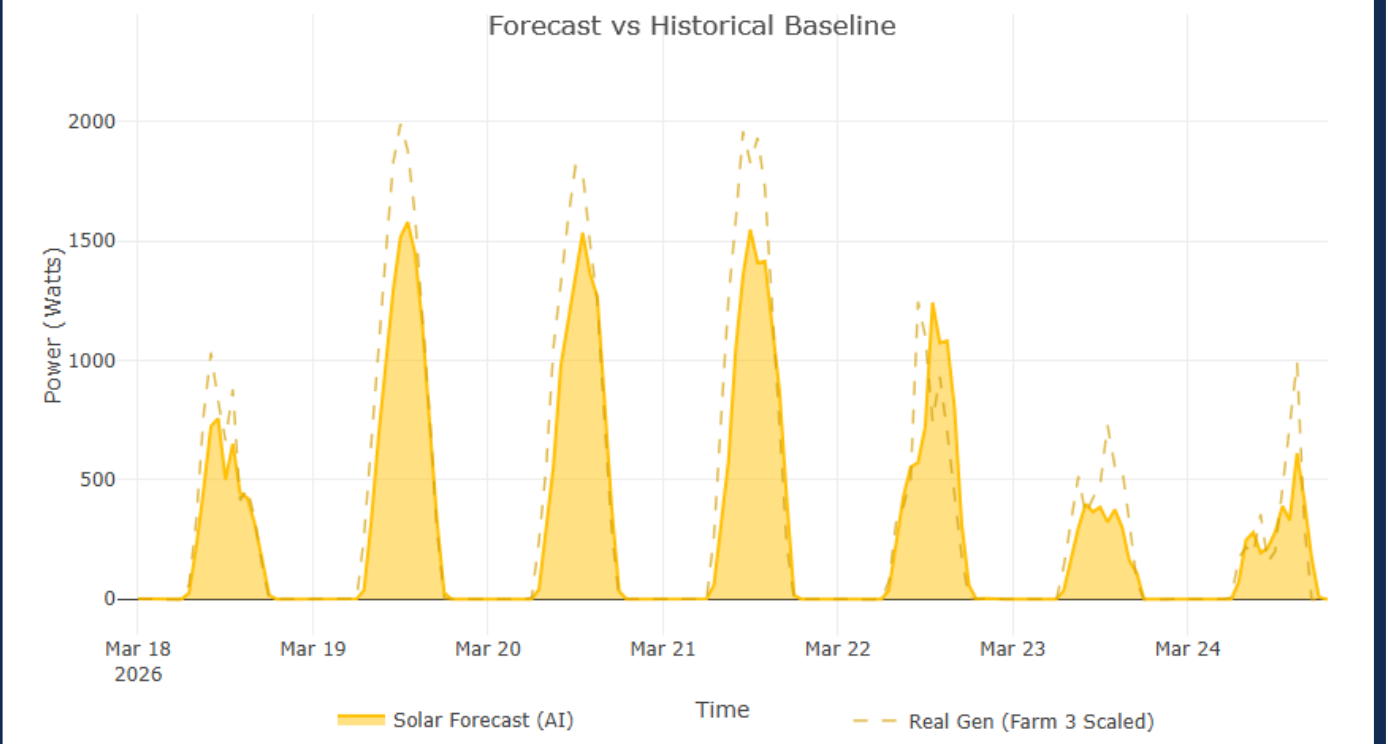


Fig 9. Predicted versus measured on farm solar generation

- Comparative Benchmarking:** Core metrics across four sites (17 to 54 WhLm⁻¹) validate the digital twin against industry baselines (41.1 to 48.9 WhLm⁻¹) [2], [3].
- A 10-month evaluation produced the most accurate comparison (47 WhLm⁻¹).

Conclusion

- This research **establishes a validated real-time Digital Twin framework**, enabling benchmarking against industry baselines while extending capability through AI-driven forecasting and anomaly detection.
- By overlaying empirical solar and wind data onto profiles of farms yet to adopt renewables, the system provides a grounded, real-time assessment of potential ROI aligned with current energy usage. This **empowers farmers** to refine operational processes and evaluate how battery or alternative energy storage solutions can optimise load-shifting to actively mitigate generation mismatches.
- Ultimately, the architecture provides a scalable mechanism to actively evaluate and **optimise farm-level renewable energy adoption**.

Future Work

- Integrate regional empirical battery, and established wind energy data for enhanced load-shifting and wind forecast simulations.
- Expand the IoT infrastructure to a farm with a robotic milking parlour.



DT Demo &
Further
Information